Assignment: ARMA Modeling and Forecasting of a Time Series

Objective

The purpose of this assignment is to analyze the persistence properties of a univariate time series using ARMA models, evaluate model adequacy, and perform forecasting exercises.

Instructions

Choose a univariate time series of your interest. Possible examples include:

- Financial data (stock prices, exchange rates, interest rates)
- Macroeconomic variables (GDP growth, inflation, unemployment)
- Commodity or energy prices
- Climate or environmental data

The dataset should contain at least 100 observations and must be downloaded from a reliable source such as FRED, Yahoo Finance, ECB, OECD, or the World Bank.

Tasks

1. **Download and prepare the data**
   - Describe the selected variable and explain its relevance.
   - Plot the time series and comment on its main characteristics.
   - If necessary, transform the series to achieve stationarity.
2. **Analyze the persistence of the process**
   - Compute and plot the autocorrelation function (ACF).
   - Compute and plot the partial autocorrelation function (PACF).
   - Based on the correlogram, discuss possible ARMA specifications.
3. **Estimate ARMA models**
   - Estimate several ARMA(p, q) models.
   - Compare the models using the Bayesian Information Criterion (BIC).

- Select the preferred specification and justify your choice.

4. **Residual diagnostics**

   - Plot the residuals of the selected model.
   - Plot the residual correlogram (ACF and PACF of residuals).
   - Perform an autocorrelation test on the residuals (e.g. Ljung–Box test).
   - Test the residuals for normality (e.g. Jarque–Bera test).
   - Discuss whether the residuals behave as white noise.

5. **Impulse Response Function (IRF) analysis**

   - Compute and plot the impulse response function of the selected ARMA model.
   - Interpret the persistence and dynamic effects of a shock.

6. **Forecasting exercise**

   - Split the sample into an estimation period and an evaluation period.
   - Estimate the model using only the estimation sample.
   - Produce:
     - Static forecasts
     - Dynamic forecasts

Submission Requirements

- Submit a concise report with a maximum length of 10 pages.
- Include all relevant tables, figures, and interpretations.